

CS513 Phase-II Project Report

Target use case U1

Identify the most common violation categories associated with failed inspections, broken down by facility type and risk level.

298,869 inspection rows preserved	974,597 atomic violation facts created	463,558 source cells rewritten	220,254 -> 0 comparable IC violations
---	--	--	--

Team79

Member	Illinois email	Primary Phase-II contribution
Sid Wanjara	swanj2@illinois.edu	Change-summary and integrity-constraint reporting; SQL review
Drew Patel	drewp4@illinois.edu	Workflow models W1/W2; conclusions; final report assembly
Shray Srivastava	ssriv5@illinois.edu	Cleaning notebook; parser; process history; cell-level change counts

Dataset D. City of Chicago Food Inspections, 298,869 rows x 17 columns, covering 2010-01-04 through 2025-10-22. Each row is one inspection, keyed by `Inspection ID`.

Report basis. All Phase-II numbers in this report come from the executed `CS513_Phase2_Cleaning.ipynb` run recorded in `output/OtherToolHistory.json`. The Phase-I report was reviewed in full and its thirteen reported missing-value counts were reproduced exactly.

Overall summary

Phase-I correctly identified that U1 was blocked by three linked problems. Violations were packed into free text, facility types were fragmented, and risk values were not a closed domain. Phase-II implemented a workflow W that cleans those dimensions without deleting inspection rows or fake missing values.

Main result

The raw table D could only group 57,819 failed inspections by their entire Violations text, creating 54,281 groups that mostly contain one record. The cleaned relations D' support a direct, ranked U1 query over 109 violation categories and 2,296 facility-risk-category cells.

What changed

Dimension	D (original)	D' (cleaned)	Effect on U1
Structure	1 flat relation; multi-valued Violations cells	Inspection relation + atomic violation relation	Violation categories can be enumerated, joined, and counted
Facility Type	521 observed categories including blanks	16 controlled facility groups	Comparable, interpretable breakdown
Risk	3 valid levels + All + blank	3 valid levels + Unknown; 167 rows flagged	Closed grouping domain with explicit uncertainty
Code meaning	Violation number interpreted without time	code_era derived at 2018-07-01	Prevents 45 pairs of unrelated categories from being merged
Completeness	3,579 failed inspections lack violation text	fail_missing_violations flag	Exclusion is explicit and reported, not silent

Quality evidence

- **All 298,869 inspection rows were preserved**; the primary key remains unique and the violation relation has zero orphaned foreign keys.
- **974,597 violation facts** were recovered from 215,525 populated cells at a 100.00% regex parse rate.
- **463,558 source cells (9.12%)** were rewritten, dominated by date formatting and safe case/whitespace normalization.
- Across the ten integrity constraints comparable before and after cleaning, violating rows fell from **220,254 to 0**. Five further checks became expressible only after the split into two related tables.
- The dataset still cannot answer U2 - which establishments caused illness outbreaks - because it contains inspection evidence, not patient, laboratory, or outbreak data.

1. Motivation, dataset, and use cases

The City of Chicago Food Inspections extract combines establishment identity, location, inspection metadata, outcome, and geographical fields in one CSV. The important conceptual relationship is one establishment to many inspections, and one inspection to zero or many violations. The original file represents the second relationship inside one semi-structured text cell rather than as rows.

1.1 Why cleaning is necessary and sufficient for U1

U1 asks which violation categories occur most often in failed inspections and how those patterns vary by facility group and risk level. D already contains the required evidence - Results, Facility Type, Risk, Inspection Date, and Violations - so cleaning is sufficient. However, the category itself is not a value in D; it is a substring mixed with inspector comments and up to 39 other violations. Direct grouping therefore treats almost every full narrative as a separate category.

U1: Identify the most common violation categories associated with failed inspections, broken down by facility type and risk level.

1.2 Control and boundary cases

Use case	Question	Cleaning implication
U0 - no cleaning needed	Count inspections by official Results value.	The seven-value Results vocabulary was already valid; it was verified and left unchanged.
U1 - target	Rank failed-inspection violation categories by facility type and risk.	Requires standardization, a 1-to-many split, code-era tracking, and clear quality flags.
U2 - cleaning not sufficient	Determine which establishments caused illness outbreaks.	Not answerable: D has no patient, lab, visit-history, or outbreak-confirmation records.

1.3 Phase-I data-quality inventory confirmed

At the start of Phase-II, the notebook re-derived every missing-value count reported in Phase-I. All thirteen matched exactly, confirming that the planned workflow still addressed the actual extract.

Field	Missing	Field	Missing	Field	Missing
Violations	83,344	Facility Type	5,264	AKA Name	2,412
Latitude	1,022	Longitude	1,022	Location	1,022
City	162	Risk	87	State	58
Zip	42	License #	18	Address	3
Inspection Type	1	Inspection ID	0	Results	0

Key distinction

Missing does not always mean incorrect. Blank Violations may be reasonable for a passing inspection, but a failed inspection with no violation evidence is a material limitation for U1. The workflow classifies and flags this condition instead of filling it.

2. Actual data-cleaning workflow W

W was implemented as a reproducible Python/pandas notebook with DuckDB SQL for before/after integrity checks. Raw columns were retained beside cleaned columns, each operation was logged, and key data rules were checked before output. The high-level outer workflow W1 is acquire and review -> profile -> clean (W2) -> verify -> document.

2.1 Executed steps

Step	Operation	Input -> output	Measured result
1	Load and profile D	CSV -> text dataframe	298,869 x 17; unique key; Phase-I counts reproduced
2.1	Normalize City	City -> city_clean	90 -> 73 observed values; 959 cells changed
2.2	Normalize State	State -> state_clean + flag	58 blanks -> UNKNOWN; 18 non-IL rows flagged
2.3	Close Risk domain	Risk -> risk_clean + flags	167 blank/All rows -> Unknown and flagged
2.4	Verify Results	Results -> result	Seven-value domain confirmed; 0 cells changed
2.5	Parse dates and types	date/type -> ISO, code_era, flags	All dates parsed; type 111 -> 101 observed values
2.6	Trim identity/address	names/address/geo -> clean forms	Case/whitespace only; no entity resolution or imputation
3	Group facility types	520 raw strings -> facility_group	16-value vocabulary; 190 nonblank rows remain Other
4	Parse Violations	packed text -> violation relation	974,597 rows; 100.00% parse rate; 110 era-aware categories
5	Classify missing values	raw + parsed -> quality flags	3,579 Fail-with-no-violation rows flagged
6	Assemble D'	clean columns + flags -> 2 relations	Row count preserved; key unique; zero orphans
7-10	Verify, answer U1, export	D and D' -> evidence artifacts	15 ICs, change summary, U1 result, 16-operation history

2.2 Parsing and category identity

Each nonblank Violations cell was split on the pipe separator. Every chunk was matched to number. `title - Comments: comment` and emitted as one row with `inspection_id`, sequence, code era, number, title, comment, parse status, and number-validity flag. Repeated violation codes were kept in the detailed record but counted once per inspection in U1 using `COUNT(DISTINCT inspection_id)`.

Unexpected domain fact: the 2018 code-book change

Chicago changed violation codes on 2018-07-01. Before the date, valid numbers are 1-45 plus 70; afterward they are 1-64. Forty-five numbers map to two different titles across eras. Therefore, violation number alone is not a safe category key; (code_era, number) is.

3. Comparison with the Phase-I plan

The Phase-I sequence S1-S5 remained intact, but implementation details changed after profiling revealed requirements that were not visible in the initial plan.

Phase-I plan	Phase-II actual	Reason for change
Python, SQL, and possibly OpenRefine; export OpenRefineHistory.json	Python 3.9 / pandas 2.3 notebook; 16-operation OtherToolHistory.json	The main operation creates a second table with date-dependent parsing. Code made the process re-runnable, testable, and diffable.
Integrity checks in unspecified SQL environment	DuckDB SQL in-process over raw and cleaned dataframes	No separate database import process was needed, and the before/after queries could be compared directly.
OR2YW-generated inner workflow	W2 drawn from numbered notebook steps and YesWorkflow annotations	OR2YW depends on OpenRefine history, which was no longer the project history source.
Violation codes assumed to be 1-44 plus 70	1-45 plus 70 before 2018-07-01; 1-64 afterward	The data showed a code-book replacement and 45 number/title collisions across eras.
Fold city spelling variants, with broad clustering considered	Fuzzy matching restricted to the CHICAGO anchor; suburbs only case/punctuation normalized	Frequency clustering incorrectly mapped valid MERRILLVILLE to misspelled MERRIVILLE.
Risk = All unresolved; blank values reviewed	All and blanks -> Unknown; risk_is_imputed flag	All does not represent one of the ordered risk levels. The decision is recorded as an assumption, not a fact.
Drew owned S3 cleaning	Shray implemented W2; Drew built W1/W2 and assembled the report	Work was reassigned during Phase-II coordination; final contributions are documented.

3.1 What remained faithful to the plan

- The target U1, the no-cleaning control U0, and the unanswerable U2 were unchanged.
- The dataset was profiled before rules were written, then categorical fields were standardized, facility types grouped, violations normalized, missingness classified, and before/after quality queries run.
- Two analysis relations were produced at the planned grains: one inspection per row and one violation occurrence per row.
- No rows were deleted. Blanks were not inferred from unrelated columns, and raw values were kept for traceability.

3.2 Deliberate scope limits

Results was not re-bucketed because U0 showed it was already clean. Names and addresses received only safe trim/case normalization; establishment entity resolution was omitted because U1 does not group by establishment. Missing ZIP codes and coordinates were flagged but not filled with guessed values. These limits keep unrelated cleaning work from introducing claims that the data cannot support.

4. Documentation of improved data quality

Quality was evaluated with paired before queries Q_U(D) and after queries Q_U(D'). A zero means the stated constraint holds; n/a means the raw structure cannot express the check. Across the ten comparable constraints, total violating rows fell from 220,254 to 0.

4.1 Integrity-constraint report

IC	Constraint	Before D	After D'	Verdict
IC1	Inspection ID is unique	0	0	Already clean
IC2	Risk is in the four-value domain	167	0	Resolved
IC3	City is nonnull and canonical upper/trim form	829	0	Resolved
IC4	No misspelled single-token CHICAGO variant	96	0	Resolved
IC5	State is a two-letter code or UNKNOWN	58	0	Resolved
IC6	Results is one of seven outcomes	0	0	Already clean
IC7	Facility type uses at most 16 groups	521 values	16 values	Improved
IC8	Every nonblank facility type maps beyond Other	n/a	190	New check
IC9	Violation facts are atomic (1NF)	215,525	0	Resolved
IC10	Violation FK references an inspection	n/a	0	New check
IC11	(era, number) determines one title	n/a	0	New check
IC11b	Number alone determines one title	n/a	45	Diagnostic
IC12	Violation number is valid for its era	n/a	0	New check
IC13	Every Fail has evidence or is flagged	3,579	0	Resolved
IC14	Inspection date is valid and in range	0	0	Already clean

IC11b is intentionally not 'fixed': its 45 exceptions show that number alone does not determine one title and support the code-era design. IC8 also makes remaining uncertainty visible: 190 nonblank facility-type rows (0.06%) remain Other rather than being forced into unsupported categories.

4.2 Fitness-for-use comparison

Measure	On D	On D'
U1 grouping	54,281 full-text groups for 57,819 failures	2,296 facility-risk-category cells
Queryable categories	None as atomic values	109 categories contributing to U1 (110 in all parsed data)
Violation detail	Up to 40 facts packed in one cell	One fact per row, keyed to inspection
Missing evidence	3,579 failures look like zero-violation cases	Clearly flagged and excluded from violation-frequency totals

5. Summary of data changes (Delta D)

The workflow rewrote 463,558 of 5,080,773 source cells (9.12%). The inspection row count did not change: 298,869 -> 298,869. The largest cell count is date reformatting, which changes representation but not the 3,985 distinct calendar days.

5.1 Cell-level changes by source column

Source -> cleaned	Changed	% rows	Filled	Distinct before -> after	Missing before -> after
Inspection Date -> inspection_date	298,869	100.000	0	3,985 -> 3,985	0 -> 0
Inspection Type -> inspection_type	63,561	21.267	0	111 -> 101	1 -> 1
Facility Type -> facility_group	37,459	12.534	5,264	521 -> 16	5,264 -> 0
DBA Name -> dba_name	22,607	7.564	0	34,146 -> 33,634	0 -> 0
AKA Name -> aka_name	21,991	7.358	0	32,506 -> 32,000	2,412 -> 2,412
Address -> address	17,887	5.985	0	20,208 -> 20,038	3 -> 3
City -> city_clean	959	0.321	162	90 -> 73	162 -> 0
Risk -> risk_clean	167	0.056	87	5 -> 4	87 -> 0
State -> state_clean	58	0.019	58	7 -> 7	58 -> 0
Results -> result	0	0.000	0	7 -> 7	0 -> 0
License / Zip / Latitude / Longitude	0	0.000	0	preserved	preserved

5.2 Structural and derived changes

Measure	D	D'
Relations	1	2
Inspection relation	298,869 rows x 17 columns	298,869 rows x 34 columns
Violation relation	Not present	974,597 rows x 8 columns
Atomic violation facts	0 queryable rows	974,597 queryable rows
Violation categories	Cannot be listed directly	110 era-aware categories

New analytical fields include facility_group, risk_level, inspection_year, code_era, is_reinspection, and n_violations. Quality flags identify missing violation evidence, imputed risk, missing facility type, missing city/state/ZIP/geo, and out-of-state records. Raw values remain available beside their cleaned counterparts.

Reversibility principle

Problematic records were flagged rather than deleted. This preserves the source population and moves exclusion decisions into explicit analysis queries, where they can be audited.

6. U1 results after cleaning

The cleaned relations support a direct join from inspections to era-aware violation categories. Each category is counted once per inspection, even when multiple comments repeat the same code. The headline categories below aggregate all facility groups and risk levels.

6.1 Most common categories in failed inspections

Era	#	Violation category	Failed inspections
pre-2018	34	FLOORS: CONSTRUCTED PER CODE, CLEANED, GOOD REPAIR, COVING INSTALLED...	19,045
pre-2018	35	WALLS, CEILINGS, ATTACHED EQUIPMENT CONSTRUCTED PER CODE...	17,854
pre-2018	18	NO EVIDENCE OF RODENT OR INSECT; OPENINGS PROTECTED...	16,413
pre-2018	33	FOOD AND NON-FOOD CONTACT EQUIPMENT/UTENSILS CLEAN...	16,202
pre-2018	38	VENTILATION: ROOMS AND EQUIPMENT VENTED AS REQUIRED...	15,156
2018	55	PHYSICAL FACILITIES INSTALLED, MAINTAINED & CLEAN	14,796
2018	38	INSECTS, RODENTS, & ANIMALS NOT PRESENT	11,380

These counts should not be compared by number across code eras. For example, number 38 has a different title before and after the 2018 code-book change. The title plus era is the safe interpretation.

6.2 Coverage of failed inspections by risk

Risk level	Failed inspections	Excluded: no evidence	Usable for U1
Risk 1 (High)	42,534	2,089	40,445
Risk 2 (Medium)	10,291	665	9,626
Risk 3 (Low)	4,968	800	4,168
Unknown	26	25	1
Total	57,819	3,579	54,240

The 3,579 excluded records equal 6.19% of failed inspections. They remain in D' and are included in general failure counts, but not in violation-frequency analyses because they provide no violation evidence.

6.3 Interpretation and limitations

- Physical-facility cleanliness and pest-control categories dominate the overall ranking, but the exact ranking varies by facility group, risk, and code era.
- The result describes patterns but does not prove cause. An inspection violation does not establish that an establishment caused an illness.
- Facility group Unknown contains 5,264 inspections; Other contains 190 nonblank but unmapped records. These categories should remain visible in dashboard filters.
- Only one of the 26 failed inspections with Unknown risk contains violation evidence, so risk-stratified conclusions for Unknown are not meaningful.

7. Findings, problems, lessons, and next steps

7.1 Findings and problems encountered

Finding / problem	Response	Lesson
The field that blocked U1 appeared well formatted but packed 974,597 facts into 215,525 cells.	Created a second, atomic violation relation.	Data-quality work often concerns structure, not spelling or nulls.
Violation-number meanings changed in 2018.	Derived code_era from inspection date and tested that each era and number maps to one title.	Domain history can be required to define a correct key.
Fuzzy city clustering proposed an incorrect correction.	Restricted fuzzy matching to the CHICAGO anchor.	Automated rules need clear limits and must be checked against exceptions.
Some failures have no violation text.	Flagged 3,579 rows and excluded them explicitly from category frequencies.	Flags preserve evidence and make limitations measurable.
OpenRefine no longer fit the critical workflow.	Used pandas plus an operation log; documented the deviation.	Tool choice should match the required data changes and record-keeping needs.

7.2 Lessons learned

- **Not cleaning is a legitimate result.** Results already satisfied its domain constraint, so changing it would add risk without improving U1.
- **Flags outperform silent deletion.** D' preserves the source row count and lets each analysis state its exclusions.
- **Before/after queries must measure the same rule.** DuckDB over both tables allowed the same constraints to be evaluated where structurally possible.
- **Some improvements make new checks possible.** Foreign-key validity, one-violation-per-row structure, and era-aware title mapping could be checked only after the data was normalized.

7.3 Practical implementation of U1

A production implementation would load both cleaned relations into DuckDB or a warehouse, save the Step 9 query as a database view, and provide filters for facility group, risk level, inspection date, and code era. The dashboard should rank categories, show the total records used, and display the Fail-with-no-violation count as a persistent quality footnote.

Next step	Purpose
Publish a versioned view keyed by (facility_group, risk_clean, code_era, violation_title)	Prevents code-era collisions and standardizes downstream analysis.
Add automated regression tests for all 15 ICs	Detects changes in the portal schema or category values.
Review the 190 Other facility records and 5,264 Unknown records with domain experts	Improves coverage without guessing categories.
Track parsing and missing-evidence rates by extract date	Separates true trends from data collection changes.
Join external illness/outbreak data only under an approved linkage design	Would be required for U2; D' alone remains insufficient.

8. Reproducibility, process history, and contributions

The executed notebook is both the cleaning implementation and the human-readable process record. Each transformation has a numbered markdown section, a one-line rationale for U1, before/after metrics, and an entry in OtherToolHistory.json. The operation history records tool versions, parameters, input/output columns, affected rows, timestamps, constraints, and the change summary.

8.1 Deliverable map

Artifact	Role in the evidence chain
CS513_Phase2_Cleaning.ipynb	Executed workflow W2; profiling, cleaning, checks, U1, outputs
Workflow.yw / Workflow*.gv / .pdf / .png	YesWorkflow annotations and rendered W1/W2 models
output/OtherToolHistory.json	Machine-readable log of 16 executed operations
queries.txt	SQL for 15 integrity constraints, U1, and profiling
output/ic_report.csv	Recorded Q_U(D) and Q_U(D') results
output/change_summary.csv	Cell-level Delta D by source column
output/derived_columns.csv	Definitions and populations of derived fields and flags
output/u1_result.csv	2,296-row facility-risk-era-category result
DataLinks.txt	Dataset metadata and locations; Box link placeholders remain to be filled

8.2 Team contributions

Member	Contribution
Shray Srivastava	Built and executed W2: profiling, categorical cleaning, facility mapping, violation parser, quality flags, two output relations, operation history, and cell-level change counts. Drafted the cleaning narrative.
Sid Wanjara	Prepared and reviewed the Delta D summary, integrity-constraint reporting, and queries.txt evidence.
Drew Patel	Built workflow models W1/W2, synthesized conclusions and next steps, and assembled and quality-checked the final report.

8.3 Reproduction

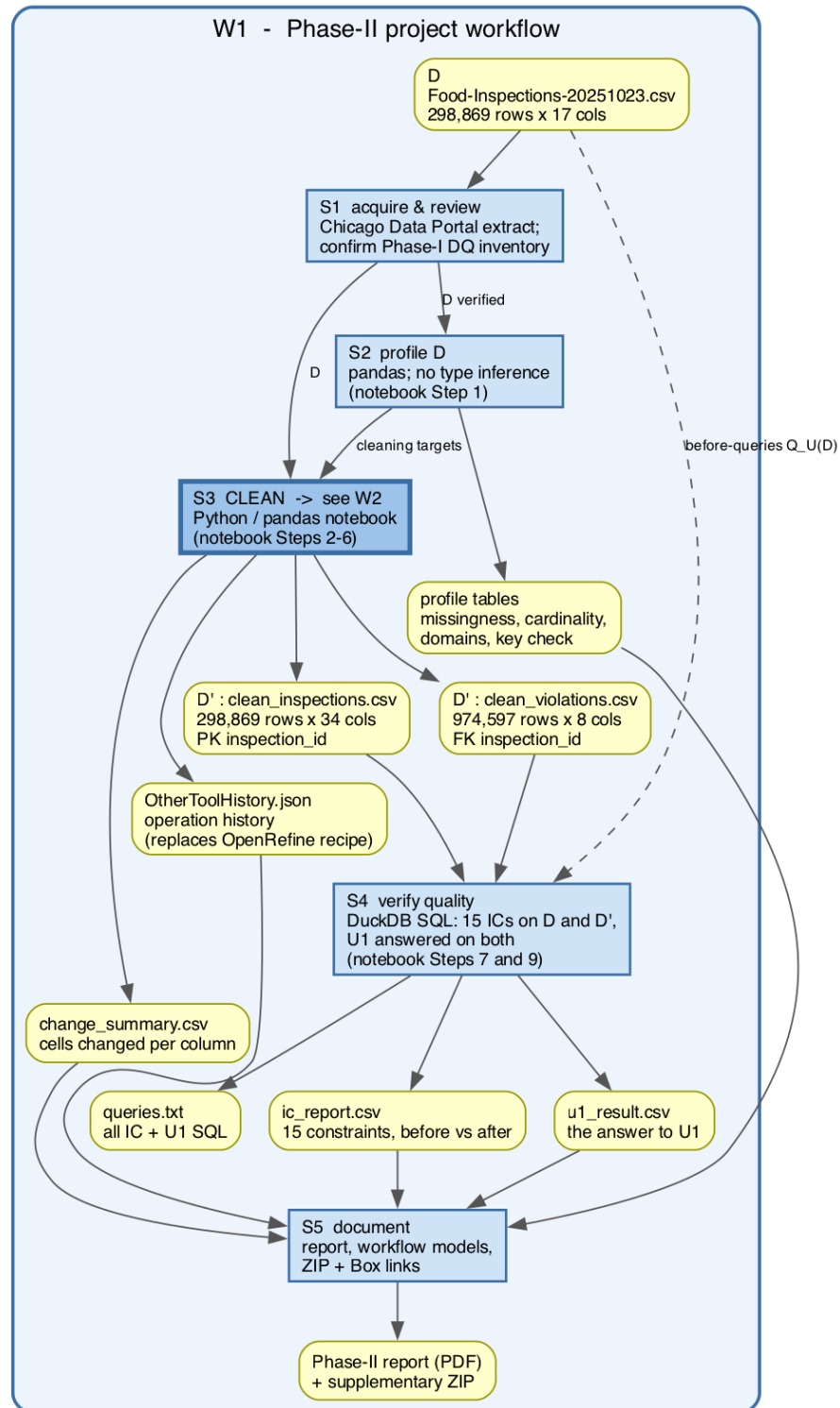
Place Food-Inspections-20251023.csv beside the notebook and run all cells with Python 3.9+, pandas 2.3+, and DuckDB 1.4+. The recorded run completed in about 24 seconds on the contributor's machine and wrote the two cleaned relations plus every supporting evidence file. Dataset links must be inserted into DataLinks.txt before the final submission package is uploaded.

Data source: City of Chicago Data Portal, Food Inspections, data.cityofchicago.org/.../4ijn-s7e5.

Appendix A. Workflow W1 - outer project workflow

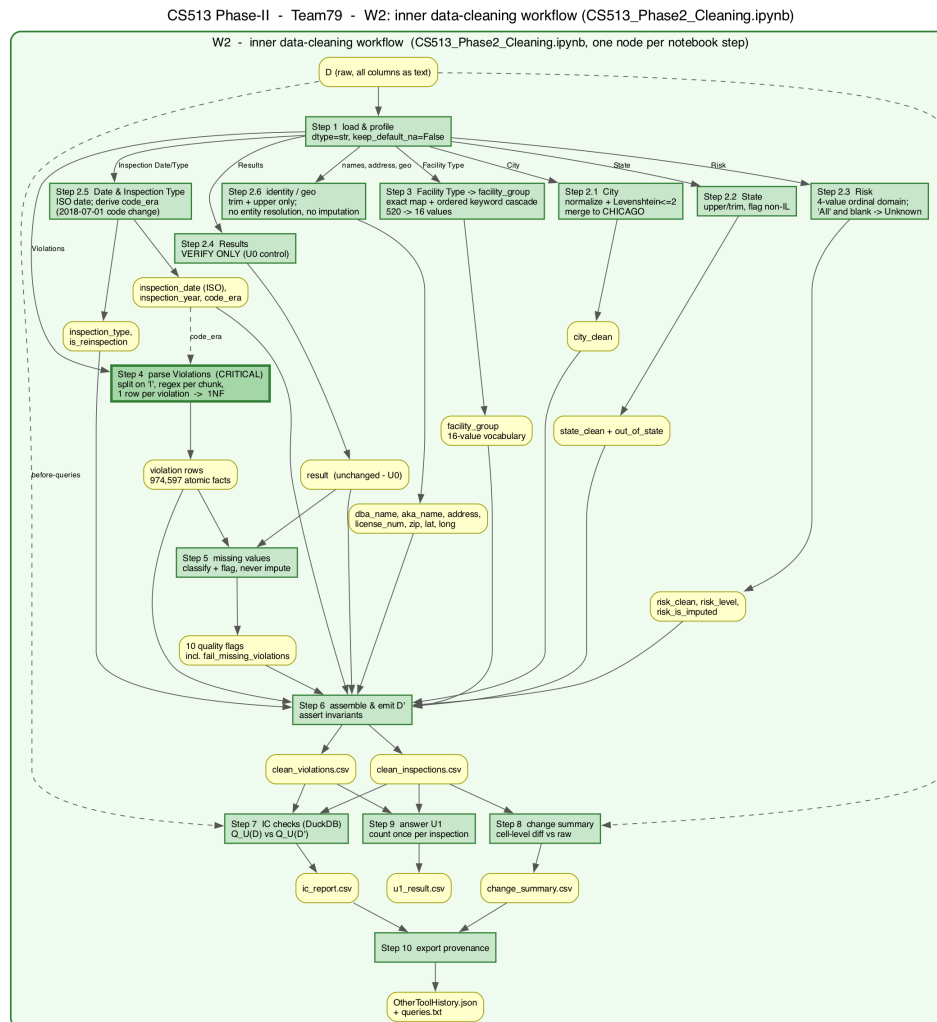
W1 connects acquisition and profiling to the inner cleaning workflow W2, then to paired quality verification, U1, process records, and report artifacts.

CS513 Phase-II - Team79 - W1: outer project workflow



Appendix B. Workflow W2 - inner cleaning workflow

W2 expands notebook Steps 1-10. The non-obvious dependency `inspection_date` -> `code_era` -> violation parser is explicit.



Reading the model

Green rectangles are processes; yellow rounded boxes are data products. Step 4 creates the atomic violation table. Step 6 checks the row count, key, and foreign-key rules before Steps 7-9 generate the quality and U1 evidence.